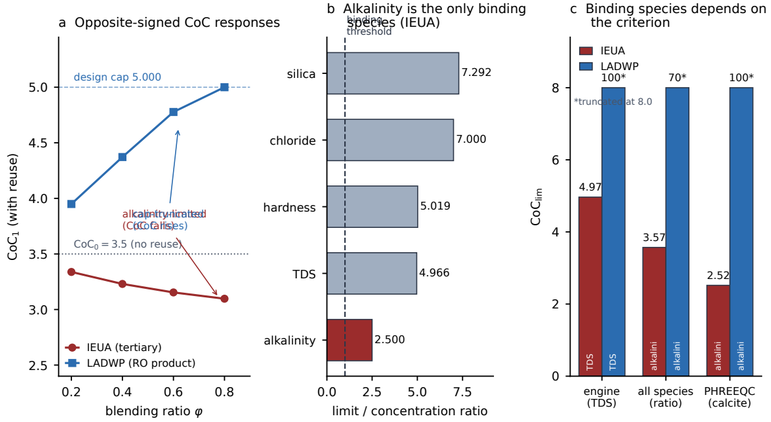

Data: A33_WP4/wp4_results.csv (a; pool_scale=1 slice, n_eff=47), A4/T6_coc_limits.csv (c), A4/T6_water_chemistry_samples.csv + A27 cited-limit tier (b); NE-2 boundary from A38_simulated-review/ne2_panel_robustness_summary.json.
n_eff = 47 city-level scenarios after the dual gate (converged_unique & not coc-floor-degenerate); IEUA n=24, LADWP n=23. phi=0 rows excluded by the gate (CoC-floor degenerate).
LIMITATION: panel b is a static headroom ratio at the cited tier, NOT a dynamic threshold (the dynamic flip point for alkalinity is ~2.615x, Fig 5). Its 'only binding species' reading holds ONLY for the cited and permissive tiers, which share an alkalinity limit of 350; under the strict make-up tier (1000/250/450/350/30) the binding species becomes silica (IEUA) and TDS (LADWP) - see Fig 5c / NE-2. Only 2 complete-species samples exist, so the opposite-signed curves in (a) are a 2-sample contrast, not a population result.

cited / permissive tier only —
strict tier (1000/250/450/350/30) ⇒ silica